
Embedded Systems International

Project Statement of Work (PD3)

Team Identifier: **G2**

Team Name (optional):

Team Member Names: **Connor Link, Erik Hjelle, Mutasim Ahmed, Benjamin Ponce-Martinez**

Submit your document as a PDF file in Canvas under the corresponding project assignment.

One of your section's lab TAs must approve of this Statement of Work by adding a comment in the Canvas assignment. The team is responsible for requesting approval promptly after submission.

Refer to the Project Requirements document before completing this Statement of Work (SOW). A statement of work is a focused concise proposal and agreement that describes work to be done. Teams should complete and submit this SOW form, which represents several parts of a statement of work, including a plan for what you are doing and how. The SOW defines the scope of your project and the approach you are taking to deliver on the goals.

Problem Statement

First, has your team reached consensus on the autonomous vehicle (AV) application you will use as the context or story for your project? All projects, regardless of application, will need to meet the same basic requirements and will be recognized for innovative features.

To define your problem, think about one or more users and their needs. Write at least one Point of View (POV) statement for your application. Follow Steps 2 and 3 in the following guide (also in Canvas as a PDF document):

[Define and Frame Your Design Challenge](#) (links to IDF webpage)

URL: <https://www.interaction-design.org/literature/article/define-and-frame-your-design-challenge-by-creating-your-point-of-view-and-ask-how-might-we>

See the Lab Project page in Canvas for the PDF file for the webpage.

Next, think about and write a few sentences that give a high-level summary of the broad mission goals for your AV application. For example:

The purpose of the AV is to provide rescue services for the US Coast Guard to recover assets lost after shipwrecks.

The AV will be capable of doing semi-autonomous traversal of a wreckage field to recover said assets.

The mission goals and user needs establish the purpose of the project and why you are working on it. Now you should translate these into a more detailed problem statement that provides a specific, concise, clear and thorough description of the context for the problem, an explanation of user needs that will be addressed, and an outline of your proposed technical approach to solving the problem.

Problem Statement: We aim to work on the final project in the context of an Underwater Search and Recovery Bot. This means considering how obstacles in the final demo can be reflected as underwater obstacles and how our robot will be able to successfully recover its target.

POV statement: the US coast guard wants to launch our Cybot underwater and allow it to navigate autonomously a designated search area in effort to find and recover some valuable assets lost. A ship sank recently, and they need the Cybot to use its integrated sensors and user input to specify its directive, then use this information to locate and recover said asset.

High-level mission goals: The purpose of the AV is to aid in the search and recovery of assets in underwater environments. The AV will be capable of using its sensors to avoid various obstacles and locate its target. The AV can also receive user input to help it work under the specific parameters of the situation it is being sent into.

Short description of the problem: By creating an underwater AV that can autonomously navigate its way around obstacles underwater, users such as the US Coast Guard can search for and recover important assets without the risk of sending a diver or the challenges faced by trying to manually navigate wreckage. The AV will be able to receive input from the user as well as make use of its onboard infrared, PING))), and open interface sensors. Using the data it collects, the AV will navigate over, around, or through obstacles it may encounter underwater as it works to locate the target designated by the user. To accomplish this, we will include movement functions that allow it to move, as well as functions to process the data collected by the sensors to identify which objects need to be avoided and which objects could potentially be the target. The robot aspires to navigate from its wreckage launch site to the valuable asset in under 20 minutes.

Problem sketch attached at bottom of document.

In addition to writing a paragraph about the problem, you are to draw a **problem sketch: a one-page sketch illustrating your solution with a user context (big picture view)**. This should show the scope of your work in relation to one or more user needs. Refer to the sample project sketches.

Design Approach

Next, consider your AV application in relation to the project requirements and the five categories by which it will be evaluated.

- 1) Functionality in relation to the AV application mission goals and user needs ^[1]_{SEP}
- 2) Mapping of functional requirements to platform components and capabilities ^[1]_{SEP}
- 3) Elements of the test field ^[1]_{SEP}
- 4) Serious incident penalties ^[1]_{SEP}
- 5) Feature bonuses ^[1]_{SEP}

In this section, you will identify and describe how you will design your application for each of these categories.

To complete the tables below, your team may want to use ideation tools, such as Lotus Blossom. Refer to the Lab Project page in Canvas and see the project ideas guidelines and Lotus Blossom worksheet as needed.

In addition to completing the tables below, you are to draw a **technical system sketch: a one-page sketch depicting a high-level technical system diagram of your proposed solution, such as a block diagram or dataflow diagram**. This should show both hardware and software modules.

The tables below are your initial proposal, and you may update these before your demonstration.

Functionality^{SEP}

Describe each of the basic functionalities required for the project in terms of your AV application. **The functionality must be specific to the problem and user(s).** Several functional statements are given in the example functional description for the Mars rover application in the Project Requirements document.

Basic Functionality	Mapping to AV Application
Cybot Communication	The US Coast Guard wants to run the rescue bot semi-autonomously and wirelessly, so we will use communication with the CyBot to facilitate issuing movement commands and fetching live sensor data to make intelligent decisions in locating the rescuee asset.
Cybot Movement	The US Coast Guard needs the bot to be able to move around self-propelled after receiving commands, so we will use movement on the CyBot to survey and explore to reach the rescuee. In case the wireless link is lost, the bot will continue to intelligently survey the wreckage field to attempt to find the lost asset.
Object Detection	The US Coast Guard mandates the robot be capable of finding potential environmental hazards like rough terrain or wildlife to make data-driven decisions about how to best instruct the robot to semi-autonomously find the rescuee. The bot should automatically determine which object has been detected and act accordingly with an individualized avoidance movement routine.
Object Avoidance	The US Coast Guard notes that while robust, the vessel can only sustain minor damage without risk to mission integrity or success. They require the robot to prevent collisions with wildlife (to mitigate unnecessary harm to creatures) and environmental hazards (like sharp rocks that pose considerable damage to the search-and-rescue vessel). Based upon the automated object detection, the bot will act differently with individualized avoidance policies.
Boundary Adherence	The US Coast Guard search-and-rescue team uses a surface-level ship to direct the underwater rescue at the target search area. Communications distance is limited, so the bot mustn't stray outside this useful range nor leave the desired search area. The bot will automatically identify these boundaries to ensure it never leaves the safe area.
Arrival at Destination	

	The US Coast Guard needs to efficiently find and recover the shipwreck's asset and safely return it to the surface. The bot should stop safely within the confines of the rescue zone after finding the asset and transmit the completion to the operator(s).
User Interface	The US Coast Guard is accustomed to using both standard desktop application interfaces as well as terminal-based CLI programs like PuTTY. These readouts are human-readable and easy to act upon to steer the robot semi-autonomously as the operators assist the bot around the wreckage field.
Base Station Control	The US Coast Guard expects the bot to “listen home” to remotely operate the search-and-rescue vessel semi-autonomously with robot guided sensing and object detecting and avoiding and manual vehicle human controls.
Other Application Specific Functionality (may be novel features for bonus points) Sounds to indicate robot status; <i>e.g.</i> , beeping when getting close to an object, tone played for successful navigation, etc.	

Mapping to Platform L SEP

Briefly describe how each of the basic platform components required for the project will be used in your AV application.

Basic Platform Components	Usage in AV Application
Open Interface and iRobot Sensors	Utilize on-board movement controls through the OI and integrated sensors like the bump, ping, IR, and button sensors to safely semi-autonomously navigate through the wreckage area to find the valued lost asset.
Interrupts	Program interrupts to optimize the code for UART/WiFi wireless communication by avoiding busy-waiting since the search-and-rescue bot has limited computational power and many other complex algorithms to run.
ADC	Convert analog input to digital input to determine how far the robot is from obstacles
Input Capture	Use input capture events on the rescue bot to trigger timer interrupts to determine sonic distance from obstacles as a backup to the IR distance measurement equipment driven by the ADC.
PWM	Control the CyBot's servo to scan specific ranges. The search-and-rescue bot must capture a 180-degree field of view to avoid damage and locate its rescuee efficiently.
UART/WiFi	Communicate wirelessly with the CyBot with search-and-rescue robot control information and sensor readouts
Other Platform Components or Modes (may be novel features for bonus points)	
Use an embedded IMU to augment the navigational capabilities of the search-and-rescue robot.	

Elements of the Test Field L SEP

Briefly describe a test field in the context of the real application (e.g., Martian terrain, city streets, etc.). Then state what each of the basic objects and other elements required for the test field represent in terms of the AV application. Draw and attach a **sketch of a possible simple test field for the lab**.

Test Field Description

Under an ocean where there are lots of underwater obstacles like boulders, small rocks, debris, caves, and kelp. The US Coast Guard identified a specific area in the ocean where a valuable asset was lost after a shipwreck. The submarine needs to navigate through the area to recover the asset.

Basic Objects and Other Elements	Mapping to AV Application Test Field
Tall objects (wide or composite)	Underwater Boulders, denoted by the large pipes. These will damage the bot so it must avoid hitting these at all times.
Short objects	Seaweed and small rocks and debris from ships or boats, represented by the short pipes that can be measured neither via PING))) nor IR. These won't damage the rescue bot, so it may lightly collide with them but navigate safely around.
Holes	Underwater caves, ravines, and overhangs – any dangerous area the bot will need to avoid, represented by the “crater” floor pieces. The bot mustn't fall into a crater.
Pillars (thin tall object)	Sea-floor kelp, denoted by the taller but smaller pipes. The robot should not make any contact with these to avoid snagging its propellers.
Out of bounds	Anywhere outside the expected search area, marked by white tape on the field. The robot should never cross outside the outer boundaries to avoid missing out on the valuable asset within the wreckage area.
Destination zone	The object or person to be rescued, denoted by a square of four pipes. The robot should stop within the confines of this square without hitting any of them. The Coast Guard has confirmed that the asset has fallen clear of most of the wreckage, so the area immediately surrounding the destination zone is free of obstacles.
Other Application Specific Elements (may be novel features for bonus points or incidents to avoid)	

Serious Incidents to Avoid and/or Novel Features (Optional)

You may have identified novel features in the tables above. Enter them in the table below and propose possible bonus points if demonstrated successfully. In addition, describe any additional serious incidents that might happen in your test field for your AV application.

Novel Features	Bonus Points
Successful navigation with IMU	10
Sound feature for robot	5

Serious Incidents	Deductions
Leaving the field	
Colliding with impermissible objects	

Sketches

Attach the following sketches to your submission. These were noted above in red.

- **Problem sketch**
- **Technical system sketch**
- **Test field sketch**

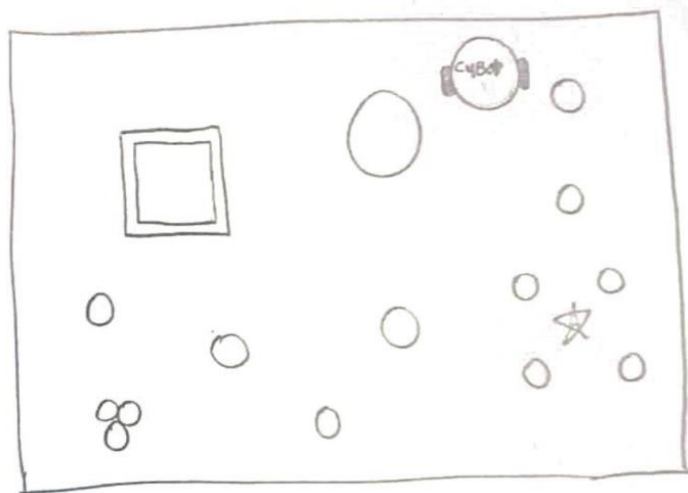
Visual communication is helpful for sharing information. There are many ways to represent your information. The problem sketch could be anything from a cartoon (informal) to a UML use case diagram (more formal). The technical system sketch is similar to what has been used/shown in class and lab. The test field sketch probably needs no explanation.

Problem Sketch



Test Field Sketch

Test Field Sketch



Technical System Sketch

